

Drew Berry

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A strong technical programmer with first-class academic credentials and a passion for AI research. During my postgraduate studies, I presented my thesis to a research group and received high commendation, with a game engine I developed later adopted by another student for their own research. I've also collaborated with the University of Essex and Oracle to deploy a large language model on Oracle Cloud Infrastructure, and I am currently developing an AI-powered music recommendation app exploring generative technologies.

SKILLS

Programming: Python (6 years), Java (4 years), SQL

Frameworks/Libraries: NumPy, TensorFlow/Keras, PyTorch, Pandas, Seaborn, scikit-learn, Matplotlib, LangChain, SQLite3, Streamlit, transformers

Tools: JetBrains suite, Git/GitHub, Jira, Google Colab, Jupyter Notebook, Oracle Cloud (OCI), Hugging Face

- Data Visualisation
- Natural Language Processing (NLP)
- Neural Network Design
- Advanced Machine Learning Techniques
- Reinforcement Learning
- Genetic Algorithms
- Predictive Analytics
- Data Cleaning and Preprocessing
- Statistical Analysis
- Technical Documentation
- Retrieval-Augmented Generation (RAG)
- LLMs

EDUCATION

MSc Artificial Intelligence | Distinction | University of Essex Oct 2023 - Oct 2024

Relevant Modules: Machine Learning, Neural Networks and Deep Learning, Intelligent Systems and Robotics, Professional Practice and Research Methodology, Text Analytics, Game Artificial Intelligence

BSc Computer Science | Honours Class I | University of Essex Oct 2020 - Jul 2023

Relevant Modules: Evolutionary Computation and Genetic Programming, Languages and Compilers, Network Security, Algorithmic Game Theory, Computer Security, Data Structures and Algorithms, Computer Game Programming, Software Engineering

PROJECTS

MSc Thesis | Python | University Project | [Link](#)

- Developed AI driving models for the TORCS racing simulation, comparing the performance of genetic and PPO reinforcement learning algorithms.
- Tuned hyperparameters and reward functions for optimal results, eliciting complex driving behaviours.
- Fixed bugs in a community-written game client.
- Presented findings to both an assessor and a research group.

AI-powered Music Recommendations App | Python | Personal Project | [Link](#)

- Designed and implemented an AI-powered web app using Streamlit that delivers personalised music recommendations from natural language prompts.
- Integrated a large language model with Last.fm API metadata to interpret user intent and generate nuanced, vibe-based track suggestions beyond simple genre or mood filters.

- Deployed a LLaMA 3.2 1B model on Oracle Cloud Infrastructure (OCI) to experiment with generative AI and Retrieval-Augmented Generation (RAG) workflows.
- Developed a custom HuggingFace-compatible embedding class and integrated LangChain orchestration to build a PDF document question-answering system, improving contextual response accuracy through advanced text preprocessing with NLTK.

- Actively participating in Kaggle competitions to sharpen my data science and machine learning skills, applying advanced techniques to real-world problems and achieving strong results in various challenges.

- Developed and compared multiple binary toxic tweet classification models, including Logistic Regression, Latent Dirichlet Allocation (LDA), and a Transformer-based deep learning model.
- Applied custom preprocessing pipelines to clean text data and improve performance; trained, saved, and evaluated models, with findings summarised in a detailed comparative report.

CERTIFICATIONS

- Gained expertise in querying, filtering, and aggregating data using SQL to extract insights from large datasets.
- Learned to use the SQLite3 library for database management, manipulation, and visualisation in Jupyter Notebooks.
- Completed an honours module on advanced SQL techniques, including stored procedures, transactions, and joins, to handle complex database operations efficiently.

- Gained expertise in using generative AI for automating ETL workflows, schema design, data pipelines, repository maintenance, and database querying, with hands-on labs and a focus on responsible AI practices and real-world applications.

EMPLOYMENT HISTORY

- Assisted customers with merchandise inquiries and selections, ensuring a positive experience.
- Applied product knowledge to provide accurate information.
- Restocked and maintained merchandise displays for a clean and organised presentation.
- Worked to a series of KPIs including upselling, display standards and stock control.

- Helped in every area of the farm, all year round.
- This includes basic equipment operation, animal handling, maintenance, basic carpentry and construction, and the use of manual and mechanical tools.

REFERENCES

Excellent academic references and professional endorsements from prior collaborators are available upon request.